

Modeling Cell-Cell Interactions in T-Cell Dependent Cytotoxicity (TDCC) Based Cancer Therapeutics

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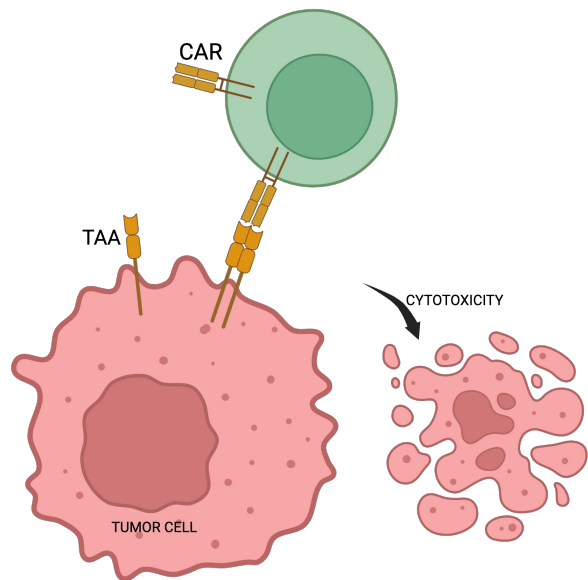
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Harnessing T Cells to Kill Cancer Cells via Tumor-Associated Antigen Recognition

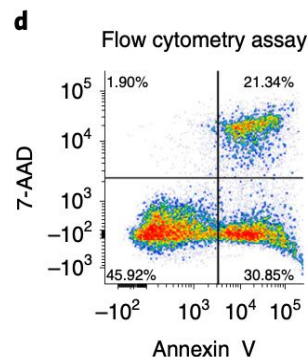
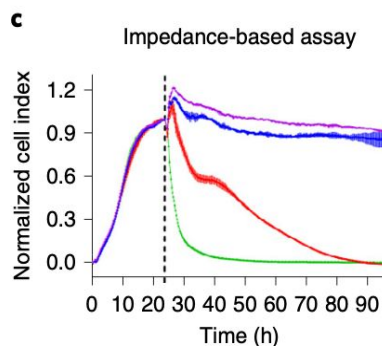
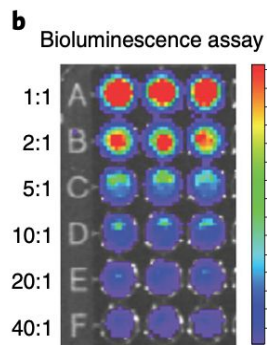
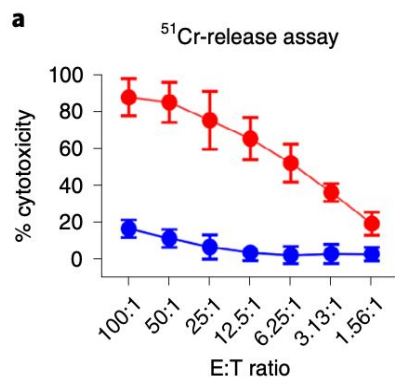
CAR T Cells (Chimeric Antigen Receptors)
TAA (Tumor-Associated Antigen)



Measuring Target Cell Killing in CAR & TCE Therapies

● Why Measuring E:T Ratios?

- High E:T ratios → maximize signal-to-noise in vitro
- In vitro conditions ≠ physiological conditions
- Modeling needed to extrapolate to real E:T scenarios



Key Questions Driving This Project

- Is E:T ratio the correct independent variable?
- Is killing invariant under fixed E:T ratios?
- Are there hidden dependencies affecting cytotoxicity measurements?



Methods

- **Agent based modeling:**
To develop intuition, “quick win”
- **ODE:**
Distribution of F across replicate wells as a function of time & N & E:T



Agent-Based Modeling Overview

“Simulate the system by simulating the parts”



Simple agents, local rules



Complex dynamics, emergent behavior

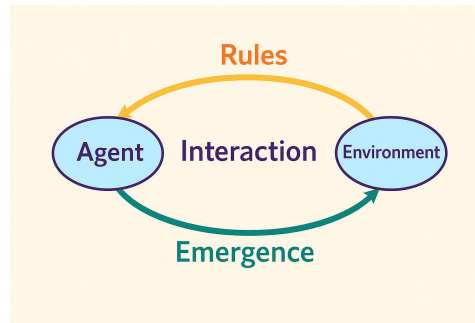
Agent-Based Modeling Overview

“Simulate the system by simulating the parts”

Agents

Environment

Actions

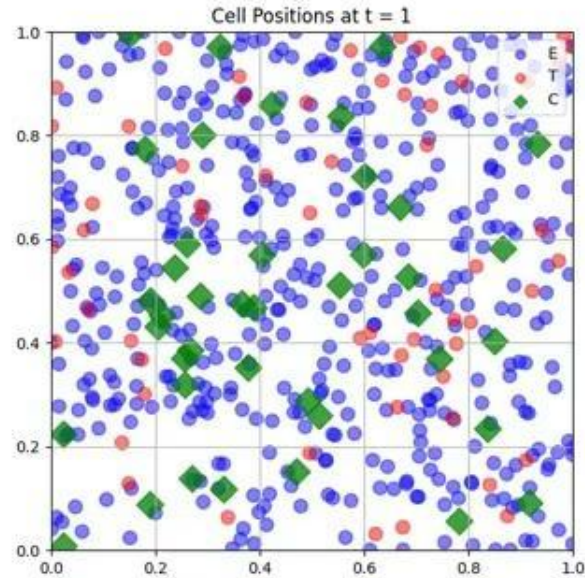
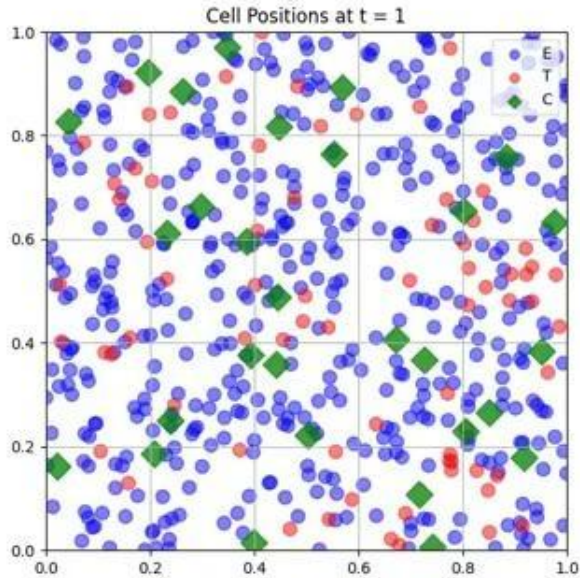


Agent-Based Model Setup

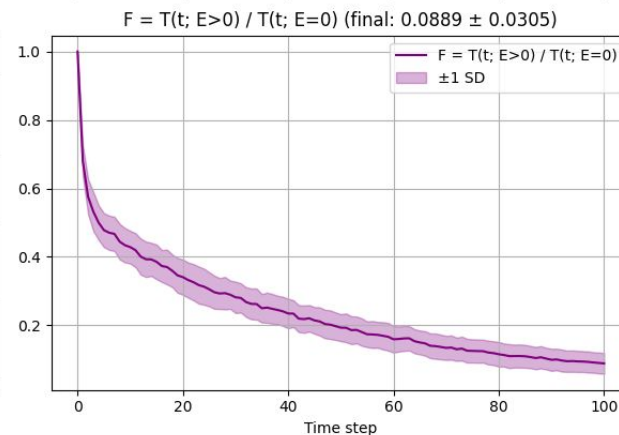
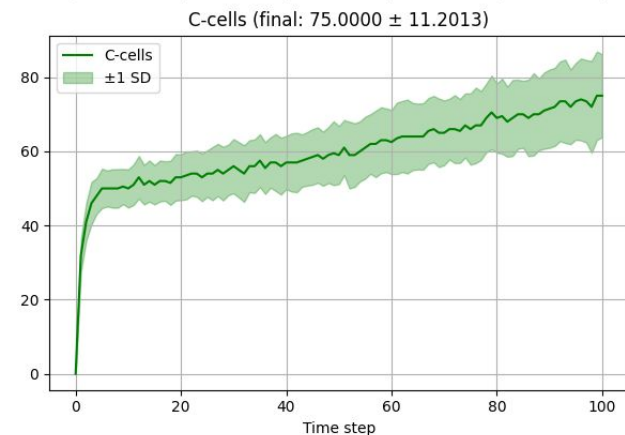
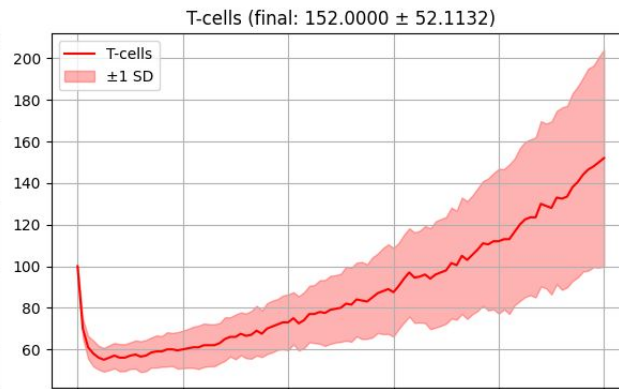
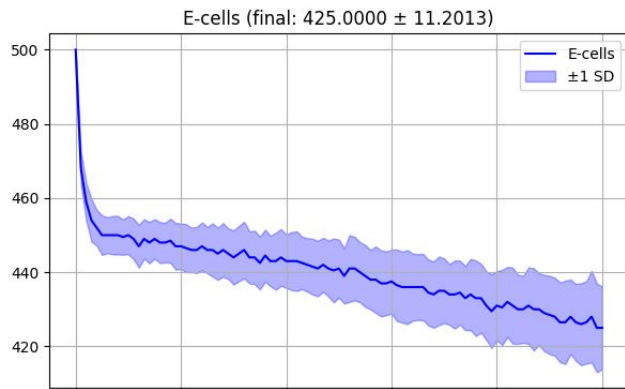
- Agents: **E**ffective Cell, **T**argeted Cell, E-T **C**omplex
- Environment: Petri dish – a '**Well**' with boundary
- Actions: Move, Bind, Kill, Split, Grow



Snapshots of ABM Simulations



Multiple Simulations Reveals Distributions



F-Metric

The F-metric compares how many T cells are left **with** E cells versus **without** them

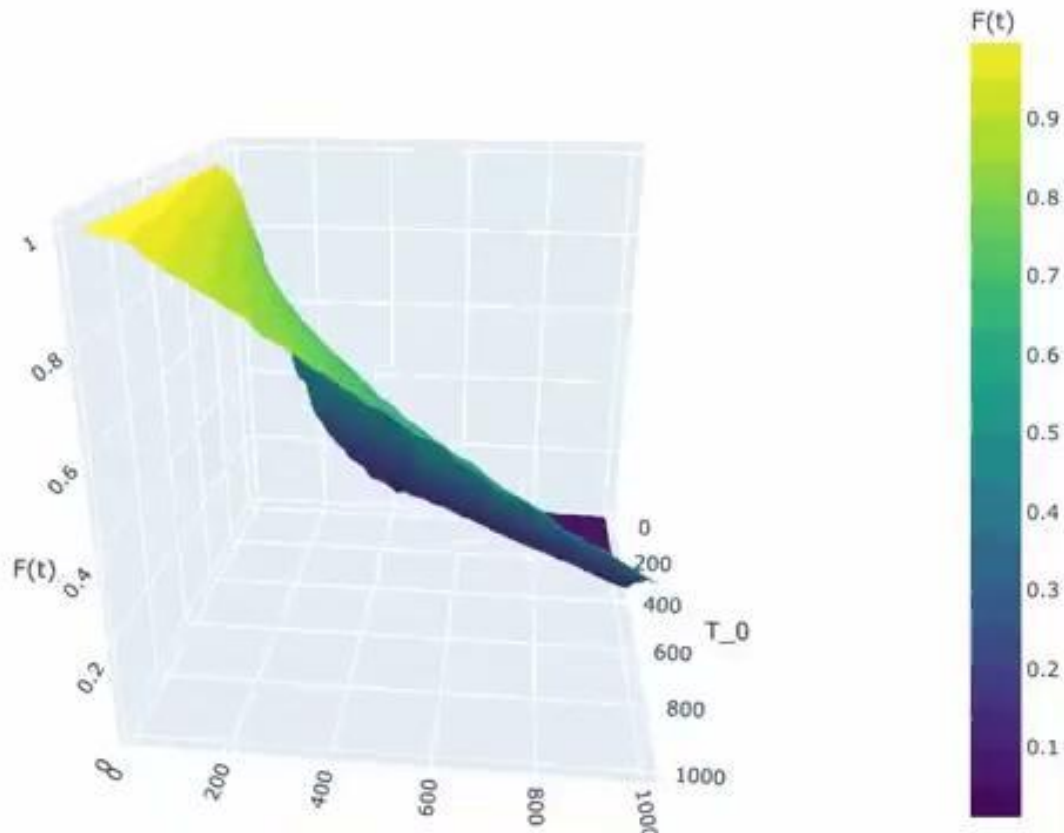
$$F(t) = \frac{T(t, E_0)}{T(t, 0)}$$

Higher F, less successful...

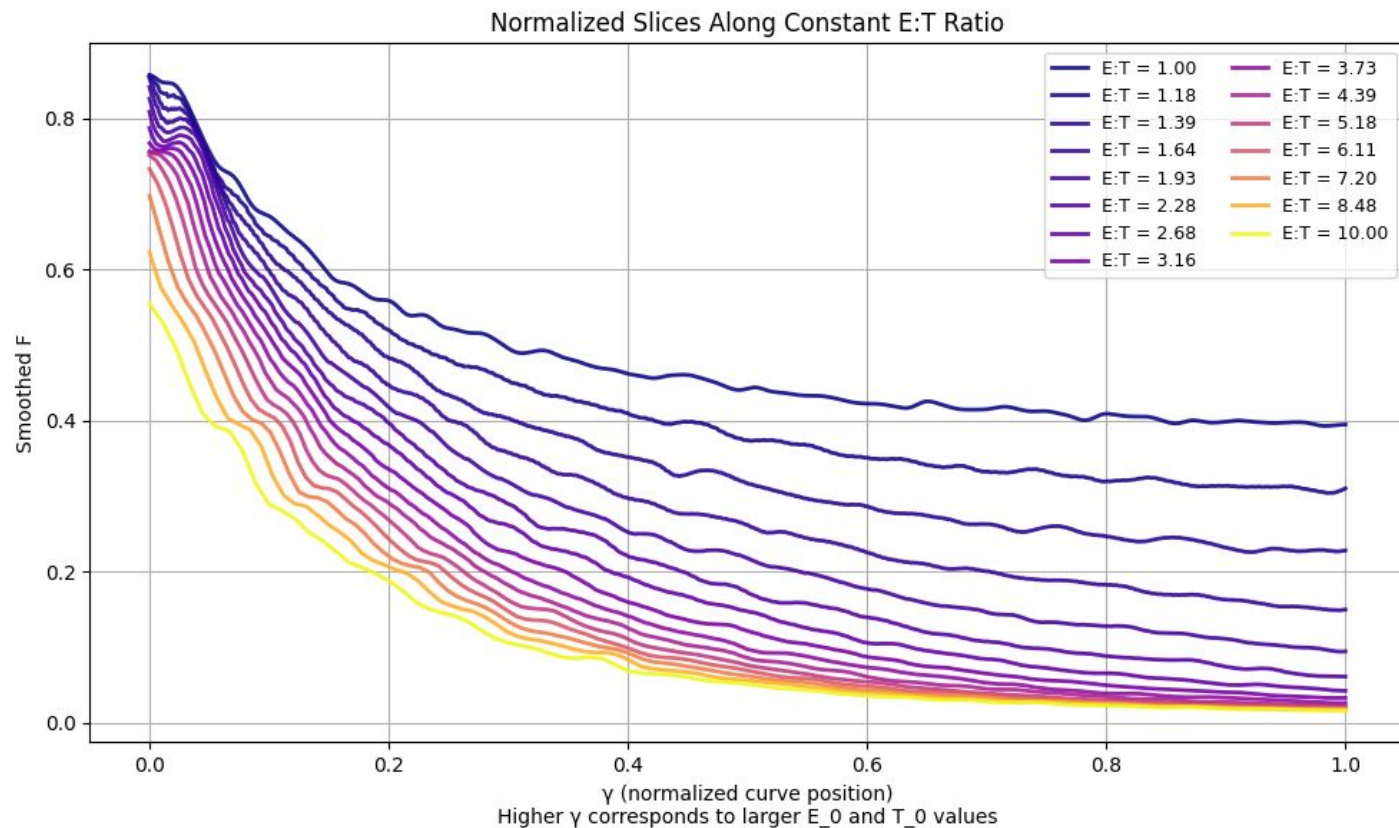
Lower F, more successful!



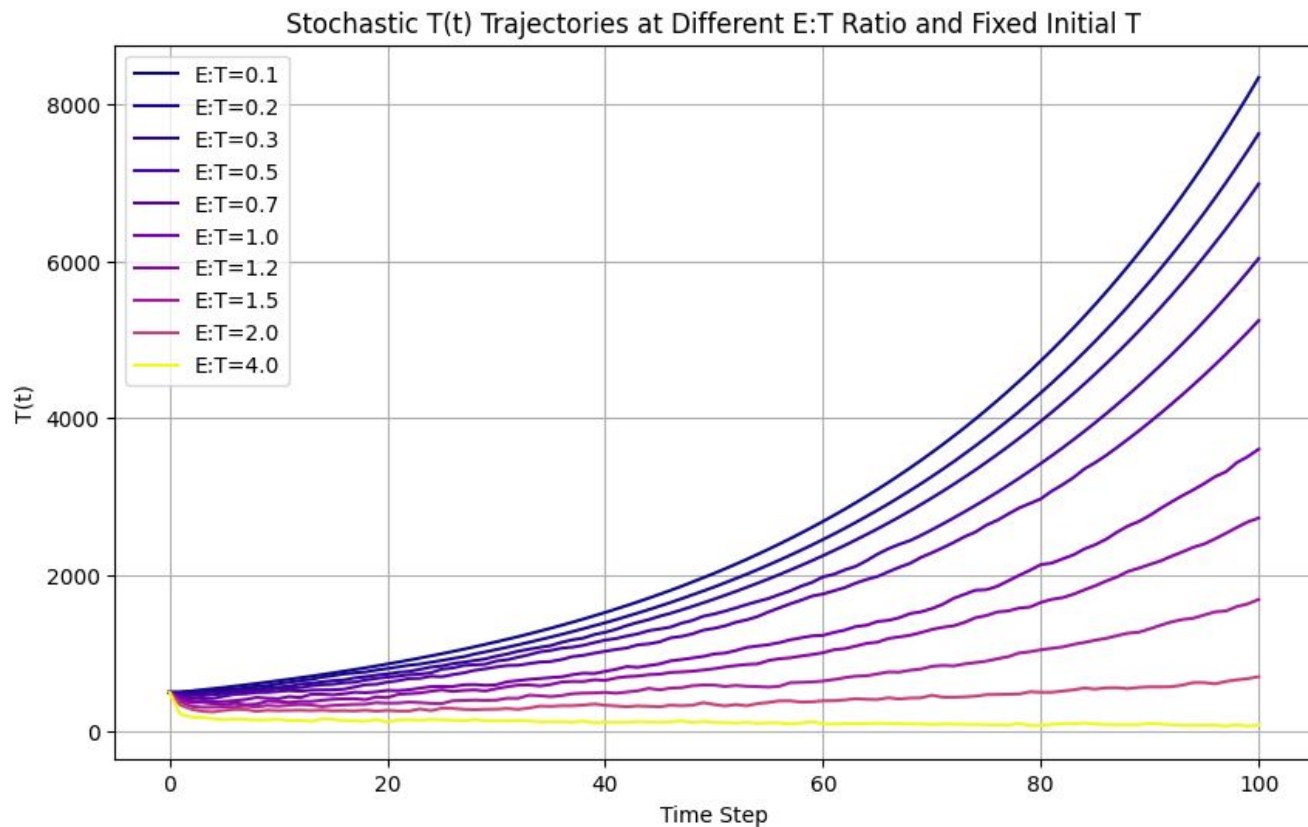
ABM F Surface



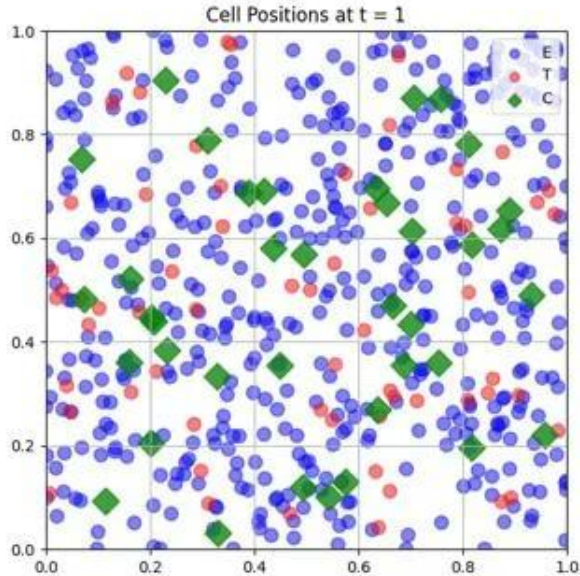
Slice Along Constant E:T Ratio



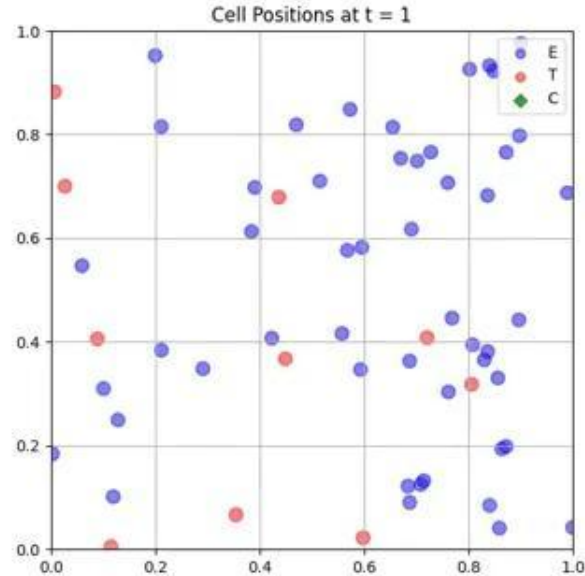
Stochastic Trajectories For Diff E:T Ratios



More E_0 and T_0, More Effective



More cells
easy to meet and bind



Less cells
effector can't find target

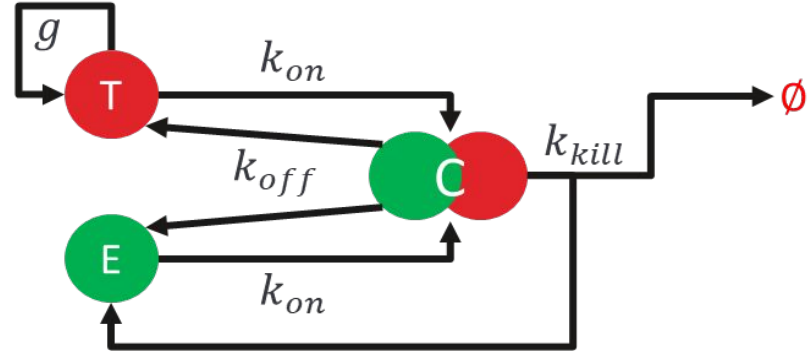


ODE models

$$\frac{dT}{dt} = gT - k_{on}E \cdot T + k_{off}C$$

$$\frac{dE}{dt} = -k_{on}E \cdot T + k_{off}C + k_{kill}C$$

$$\frac{dC}{dt} = k_{on}E \cdot T - k_{off}C - k_{kill}C$$



Cellular ‘enzymatic reaction’ where E is the ‘enzyme’ and T is the ‘substrate’

k_{on} - Binding rate constant

k_{off} - Unbinding rate

k_{kill} - Killing rate

g - Growth rate

ODE Approximation

- At the steady state, from the third equation above

$$C = \frac{k_{on} E \cdot T}{k_{off} + k_{kill}}$$

- Then, equation one and two is simplified to obtain

$$\frac{dT}{dt} = gT - \left(k_{on} - \frac{k_{off} k_{on}}{k_{off} + k_{kill}} \right) E \cdot T \quad \frac{dE}{dt} = 0 \quad \Rightarrow \quad E(t) = E_0$$

Hence,

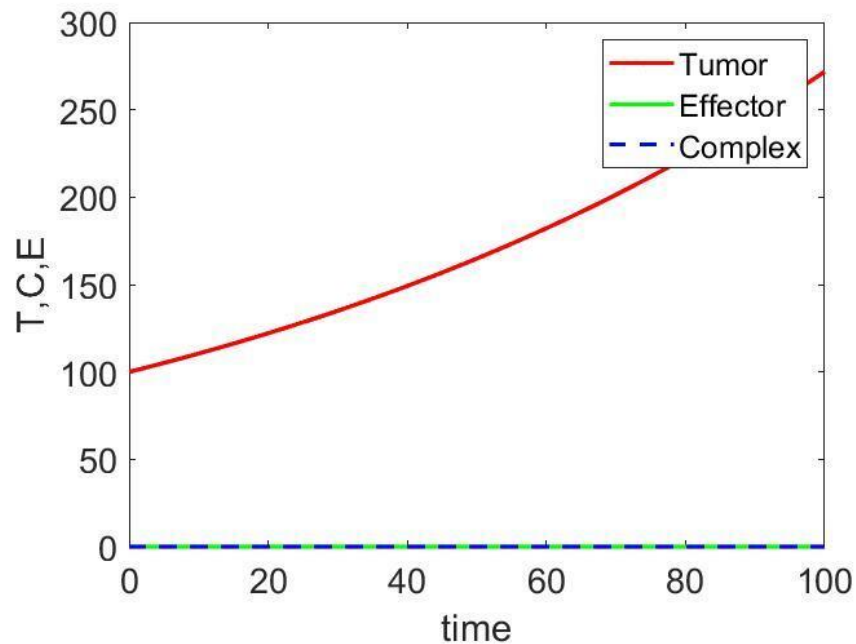
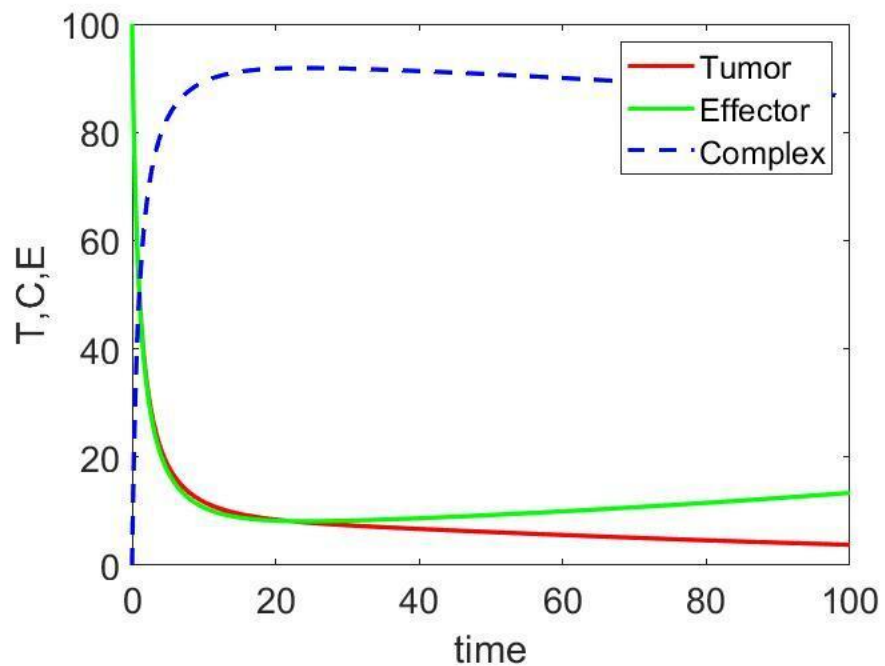
$$T = T_0 \exp \left[\left(g - \frac{k_{on} k_{kill}}{k_{off} + k_{kill}} E_0 \right) t \right]$$

- The fraction of surviving target cells at t_f

$$F = \frac{T(t_f; E_0)}{T(t_f; 0)} = \exp \left(- \frac{k_{on} k_{kill} E_0}{k_{off} + k_{kill}} t_f \right)$$

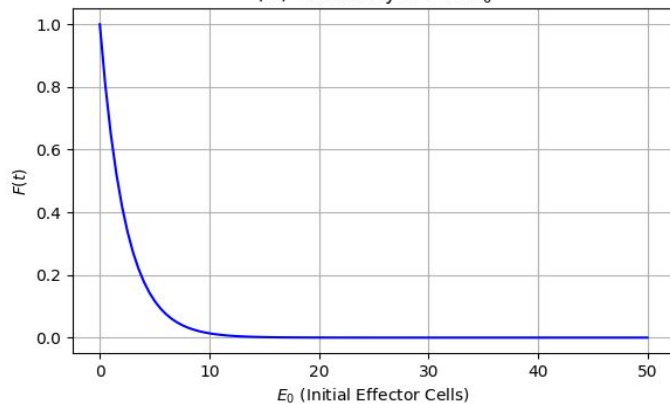


Numerical Simulations

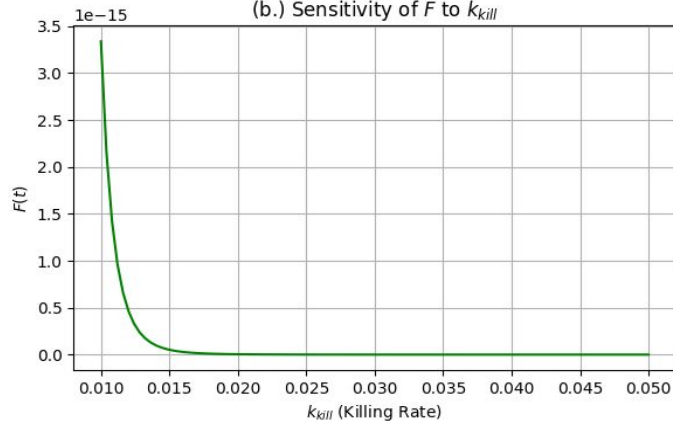


Sensitivity Analysis

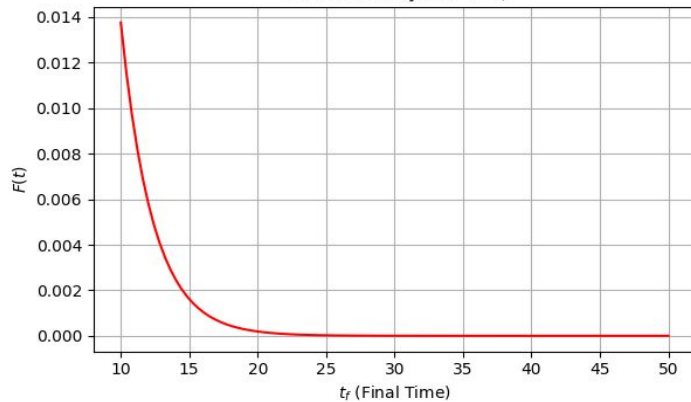
(a.) Sensitivity of F to E_0



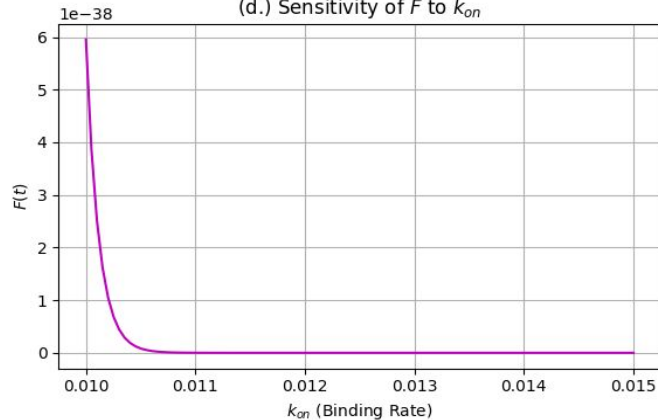
(b.) Sensitivity of F to k_{kill}



(c.) Sensitivity of F to t_f

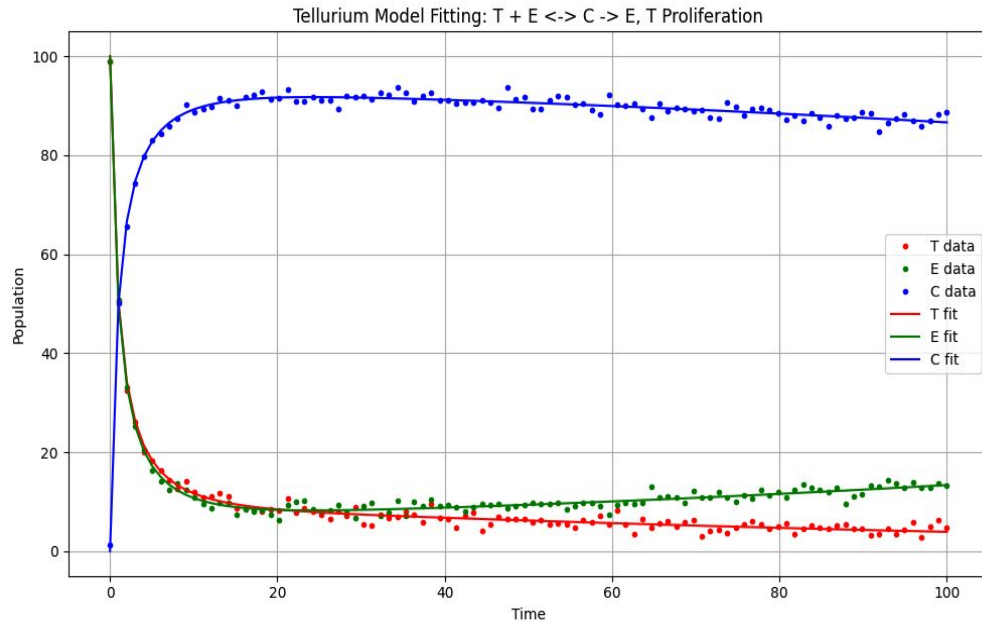


(d.) Sensitivity of F to k_{on}



Fitting Data

Goal: Estimate model parameters from observed E, T, C data



ABM Fitted:

ODE Fitted:

$r = 0.02841$

$m = 0.04759$

$k_{on} = 0.67462$

$k_{on} = 0.00997$

$k_{off} = 0.00543$

$k_{off} = 0.00515$

$k_{kill} = 0.00205$

$k_{kill} = 0.00194$

$g_E = 0.00032$

$g_E = 0.0$

$g_T = 0.01160$

$g_T = 0.00941$



Next Steps

- ODE Master Equation
- More realistic ABM setup



Thank you !

